A-A-53186
21 September 1987

COMMERCIAL ITEM DESCRIPTION

BANDAGE, CAST, FIBERGLASS RESIN IMPREGNATED

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers lightweight, rolled, resin impregnated fiberglass bandages suitable for forming orthopedic casts. The bandages shall be furnished in the following nominal sizes, as specified:

Size 1 - 2 inches by 4 yards
Size 2 - 3 inches by 4 yards
Size 3 - 4 inches by 4 yards
Size 4 - 5 inches by 4 yards

Salient characteristics:

Material. Materials used in the manufacture of the bandages shall be of high quality and suitable in all respects for the purpose intended. The gauze fabric shall be a knit fiberglass backing of warp chain loops and straight filling threads. The impregnating material used in the manufacture of the bandage shall be a water activated urethane prepolymer resin composition. The resin composition shall cure upon exposure to water to provide a hardened cast bandage.

Construction. The bandages shall consist of a knit fiberglass fabric impregnated with the specified resin and when activated by water the bandages shall be moldable to produce a radiolucent hard coated cast with a non-dusting surface. The resin shall be uniformly spread and firmly bonded to the fabric. The finished bandages, when used in accordance with the manufacturer's instructions, shall be suitable for forming orthopedic casts for immobilization purposes. The finished bandages shall appear white (with a tint due to the urethane resin) and shall be furnished as a neatly rolled bandage. The long edges of the rolled bandages shall be parallel with a selvage edge, free from raveled threads.

The rolled bandages shall be furnished as a roll, in one continuous length of material. Each bandage shall be evenly rolled on a rigid non-absorbent core having an inside diameter of 5/8 to 3/4 inch. The moisture sensitive rolled bandages shall be individually protected in a heat-sealable laminated packaging material.

AMSC N/A FSC 6510

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The laminated material shall consist of a layer of aluminum foil sandwiched between an inner layer of heat-sealable plastic polymer film and an outer layer of plastic polymer film which protects the aluminum foil from punctures, abrasion and flexural fatigue cracks.

Dimensions. Bandages shall conform to the dimensions listed in Table I for the nominal size.

TABLE I. Dimensions.

(1) Nominal Size*		(2) Minimum Width	(3) Minimum Length
Width - inches	Length - yards	inches	yards
2.0	4.0	1.9	3.9
3.0	4.0	2.9	3.9
4.0	4.0	3.9	3.9
5.0	4.0	4.9	3.9

*Nominal size of bandage is the minimum average size of the applicable bandage when sampled and examined as specified below. Width and length of individual bandages shall be no less than the values listed in columns 2 and 3 for the nominal size.

Cast strength. The bandages shall meet the cast strength of Table II for the applicable bandage size when tested as specified below.

TABLE II. Cast strength.

Hardening time, minutes	Compression, pounds/inch of bandage width, minimum
5	2.5
30	15.0
1440	45.0

The hardening times of 5 minutes correlates with the setting time; 30 minutes corresponds to the initial weightbearing in walking casts; and 1440 minutes (24 hours) is the ultimate strength of the cast material.

Water penetration. Bandages shall show uniform and complete water penetration, with no visible dry spots or areas, when tested for cast strength.

Cyclic fatigue strength. The cyclic fatigue strength shall be less than one inch of deformation at 100,000 cycles when tested as specified below.

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Porosity. The cast bandage shall exhibit a minimum transmission of 3.5 grams of water per week for a two inch wide bandage and 5.0 grams of water per week for a three inch wide bandage at 75°F and 55% relative humidity, when tested as specified below.

Elongation. The elongation of a section of non-immersed finished bandage shall not be less than twenty-five percent when tested as specified below.

Interlayer lamination strength. The cast bandage shall require a minimum force of six pounds per inch of bandage width to delaminate or tear the bandage, when tested as specified below.

Accelerated shelf life. The force to unwind a packaged finished bandage shall not be greater than 2.5 pounds per inch of bandage width after 4 weeks at 150°F, when tested as specified below.

Package integrity. The true failure rate of the laminated pouch containing the finished bandage to water leakage, shall be less than 2.0% $\pm$ 0.5% with a 90% confidence level when tested as specified below.

Irritation. The bandages shall be non-irritating and shall not cause thermal burns to the normal skin when used as directed. The bandages shall be free from harmful materials and shall have no objectionable odor.

Instructions for use. Instructions for use shall be furnished for the bandages and shall be printed on the unit package or box, or, as an alternate, on a separate sheet of white paper. The printing shall show no offsetting, smearing or bleeding and shall be clearly legible.

Lot or batch number. Each foil laminated package shall be marked with a lot or batch number and date of manufacture.

Shelf life. The finished packaged bandages shall comply with requirements of this specification for no less than 18 months from date of manufacture, when stored at 36 to 46°F.

Delivery. Not more than two months shall have elapsed from the date of manufacture of the bandages to the date of delivery to the Government.

Workmanship. Bandages and their enclosing laminated foil pouches shall be free of defects which may detract from their appearance or impair their serviceability. Bandages shall be furnished clean.

Unit. Package (PG). One package containing ten rolls of bandages, one size only, as specified, constitutes one unit. The bandages shall be supplied in sealed packages that prevent moisture penetration or other deterioration of the bandages prior to the one year expiration date.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the raw materials, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Manufacturer's protocol. A manufacturer's protocol, that lists the inspection characteristics with specification requirements, and which states that inspection results comply with said requirements, is acceptable, providing the actual inspection results are made readily available to Government representatives upon request.

Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified in Tables III, IV, and V.

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Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in Table III except for the minimum average dimensions. Unit of product for examination purposes shall be one bandage from each lot or batch number.

TABLE III. Inspection and acceptable quality levels.

Type defect	Inspection level	AQL (percent defective)
Major	S-2	1.5
Minor	S-2	4.0

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and Table IV. Where feasible the same test unit may be used for determination of two or more test characteristics. Acceptance number shall be zero for all sample sizes. The unit of product for end item testing shall be one unit of issue from each lot or batch number.

TABLE IV. Testing Criteria.

(1) Characteristics	(2) Minimum Sample Size	(3) Inspection level
Cast strength	5	S-2 2/
Water penetration	5 1/	S-2 2/
Cyclic fatigue strength	5	S-2 2/
Porosity	5	S-2 2/
Elongation	5	S-2 2/
Interlayer lamination strength	5	S-2 2/
Accelerated shelf life	50	S-2 2/
Package integrity	--	S-1

1/ Water penetration evaluated on same specimens used for cast strength.

2/ One bandage shall be tested from each unit of product (issue) taken in accordance with inspection level, provided no less than the minimum sample indicated in Column (3) shall be taken where less is indicated in MIL-STD-105.

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Examination. The bandage shall be examined for defects, including, but not limited to, those listed in Table V.

TABLE V. Classification of Defects.

Categories	Defects
Major	
101	Long edges not parallel.
102	Bandages not of size specified.
103	Center core not present.
104	Surface not hard coated and not non-dusting.
105	Bandages not free of grease spots, dirt, or stains.
106	Raveling with free threads.
107	Objectionable odor present.
108	Resin not uniformly spread on fabric.
109	Resin not firmly bonded to fabric.
110	Bandages not of one continuous length.
111	Protective laminate packaging envelope punctured, broken, or cracked.
112	Rolled bandage hardened in protective envelope.
Minor	
201	Bandages not neatly rolled.
202	Instructions not as specified.
203	Bandage color not as specified.
204	Width and length dimensions not as specified.
205	Center opening dimensions not as specified.

Tests.

Standard testing conditions. Testing shall be conducted in a controlled environment with the temperature at 75°F ± 3°F and a relative humidity at 55% ± 5% on bandages wrapped and ready for shipment by removing envelope and testing as specified. Equipment and material shall be conditioned at this temperature and humidity at least ten hours prior to testing.

Cast strength. This test shall be used to determine the strength of a 2 inch inside diameter cylindrical ring when compressed diametrically at a specified test time. Typical test times shall be 5 minutes, which correlates with the setting time; 30 minutes which corresponds to initial weightbearing in walking casts; and 24 hours (1440 minutes), which is the ultimate strength of the cast bandage. While wearing protective rubber gloves (examination or surgical), a rolled bandage shall be removed from its laminated foil pouch and completely submerged in 80.0°F ± 1°F water and the time is recorded.

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The roll is removed form the water after 30 seconds and shaken gently, without squeezing, to remove excess water. A piece of the rolled bandage 40 inches long is unwound, examined for water penetration and cut off. One end is placed against a solid mandrel of two inch outside diameter, which has been covered with a section of two inch tubular synthetic stockinet used to construct casts; then the mandrel is rotated one revolution to form a one layer ring. Using a clamp that is wider then the bandage, the specific weight is attached to the loose end of the bandage:

Nominal width, inches	Total weight with clamp, pounds
2	1/2
3	3/4
4	1
5	1-1/4

With the weight hanging freely, five more layers are uniformly wound over the first one so that no single layer overlaps any of the others by more than 3/16 inch. The excess bandage is cut off and the end is gently smoothed down. A total time of 30 seconds to wrap one six-layer ring should not be exceeded. A second ring from the same roll is wrapped in a similar manner. The first ring from a roll may be used for any of the tests but the second ring may be used for only the 24-hour test. The specified test times are measured from the time of immersion in water to the initiation of compression. A three point bend test shall be performed with an Instron Model 1122, Chattilon Model USTM or equivalent apparatus utilizing a fixture which supports the ring on the bottom as illustrated in figure 1. A blade at least as long as the cylinder shall be used to compress it. The test fixtures are attached to the upper and lower plates of the compression testing apparatus and aligned to be parallel to one another. The cylindrical ring to be tested shall be placed on its side in the fixture so that the seam rests between the two support bars. The upper fixture shall be lowered until the bar is in contact with the specimen and no load applied. Compression shall be initiated at a rate of two inches per minute. The strength shall be recorded as the peak value reached during compression and the average of five samples is reported.

Fatigue strength. Ten cylindrical rings shall be prepared as in the test for cast strength. The samples shall be cured at standard test conditions for one week.

Five of the ten rings shall be compressed diametrically between flat platens of an Instron Model 1321, 1331, MTS Model 812 or equivalent fatigue tester. The yield strength shall be recorded for each ring and the mean of the five values determined.

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The preload and compressive load conditions for each fatigue sample are 4.5% and 45.0% of the mean yield strength. The cylindrical ring to be fatigued shall be placed horizontally between two flat platens whose diameter is greater than the length of the cylinder. The preload and compressive load conditions shall be set as determined above and cycling begun with a haversine function (a non-zero mean) at 1.2 Hertz. The load versus deformation loop shall be recorded at each 10,000 cycle interval, the final deformation after 100,000 cycles and the mean value of the five samples reported.

Porosity.

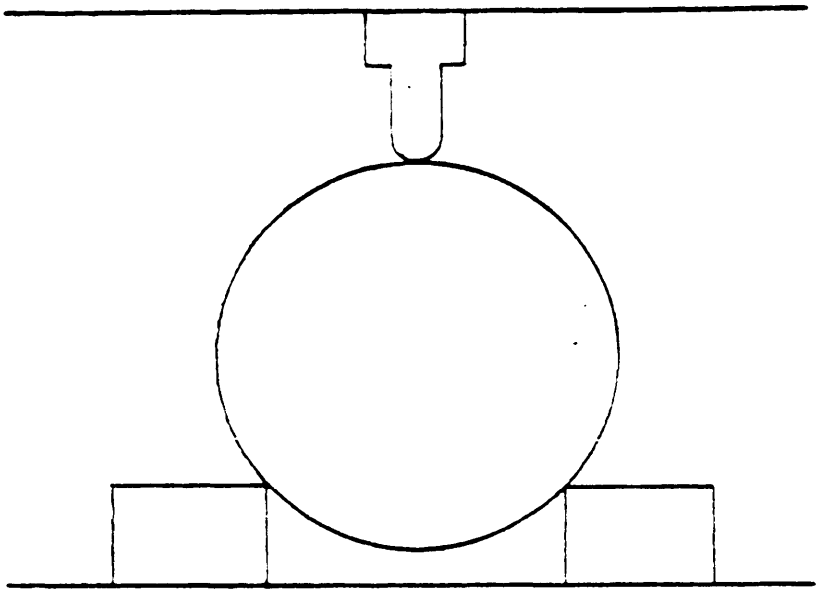
Five cylindrical rings from the same width of bandage shall be prepared as described in the test for cast strength. Bandages wider than 3 inches shall be cut to a 3 inch width. The samples shall be cured at standard tests conditions for one week. The stockinet shall be removed from inside of each ring. One end of each ring shall be sealed to a polystrene petri dish or equivalent plastic moisture barrier, with a bead of silicone sealant. A 30 ml beaker containing 30 grams of water shall be placed inside each ring on the surface of the petri dish. The top of each petri dish shall be sealed to the top of each ring with another bead of silicone sealant. The sealant shall be cured for 24 hours. The samples shall be placed in a constant temperature and humidity environment and allowed to equilibrate for one day. The initial weight of each sample shall be noted. The samples shall be weighed periodically (daily, weekly) depending on temperature and humidity conditions, to have a minimum of five readings taken before the water has completely evaporated from inside of the sample. The mean value of five readings for each sample shall be recorded and the average rate of water vapor transmission for the five samples reported as well as the temperature and humidity.

Elongation. The elongation of the bandage material under a small load is a measure of the conformability of the tape. With hands protected by rubber gloves (examination or surgical), a fresh roll of bandage material shall be removed from its package and a section approximately seven inches in length shall be cut without stretching it. The sample shall be carefully clamped between the grips on an Instron Model 1122 or equivalent extension load frame apparatus which has been set to a gauge length of five inches. The crosshead and chart recorder shall be actuated to travel at a rate of five inches per minute. The test shall be terminated at a load of ten pounds. The elongation at five pounds of load shall be calculated according to the following equation:

$$\% \text{ Elongation} = \frac{\text{Chart travel (inches from start to five pounds)}}{\text{gauge length (five inches)}}$$

The average of five samples shall be reported.

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Scale

1.0 Inch

Figure 1. View of three point diametrical compression fixture with 2" i.d. cylinder in place.

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Interlayer lamination strength.

Five cylindrical rings shall be made as specified in the test for cast strength, except an additional six inch tail of bandage shall hang freely from the ring.

After the rings have cured for 30 minutes, they shall be placed one at a time in a fixture as shown in figure 2. The ring shall be mounted on the fixture so that it can freely rotate and the fixture shall be attached to the moving crosshead of an Instron Model 1122 or equivalent. The tail shall be clamped by the stationary grips. The crosshead shall be actuated to travel at a rate of 20 inches per minute for a distance of 20 inches or until the cured bandage tears.

The delamination strength shall be the average force required to unwrap the bandage over the 20 inch distance. If the bandage tears during the delamination test, it shall be reported as such without a value. The average of five determinations shall be reported.

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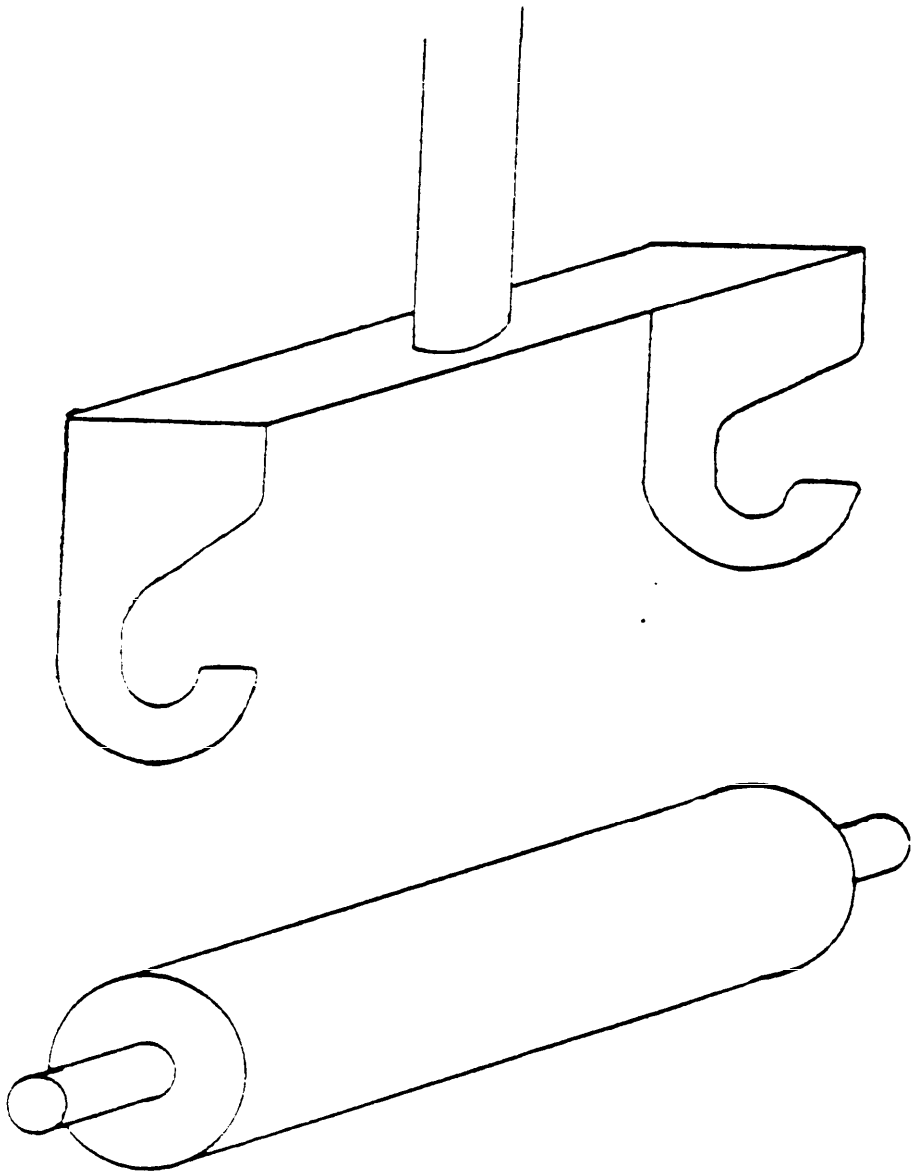


Figure 2. Fixture for lamination test.

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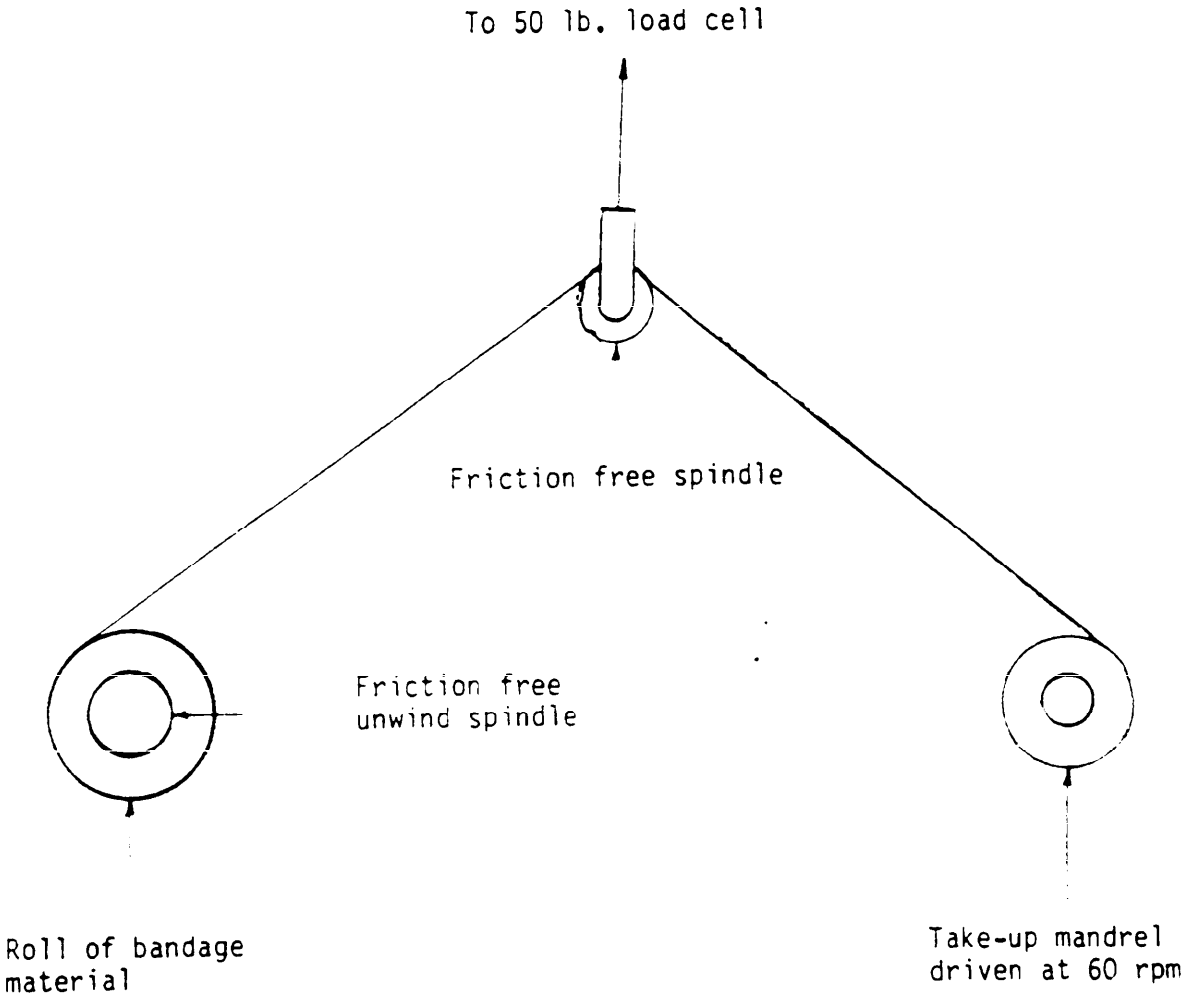


Figure 3. View of roll unwind fixture with bandage roll in place.

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Accelerated Shelf life. Fifty bandage rolls shall be sampled from the same lot and date of manufacture which are less than two months old. Five rolls shall be reserved for immediate testing and the remaining forty-five rolls shall be placed in an oven controlled at 150°F ± 5°.

Five rolls shall be removed from the oven each week and allowed to equilibrate to the standard testing conditions before testing. A fifty pound load cell shall be used in an Instron Model 1122 or equivalent apparatus. The fixture shall be as shown in figure 3. The roll of bandage material to be tested shall be mounted on a friction free unwind spindle, the leading end shall be gently draped over another friction free spindle which is fixed to the load cell and it shall be attached to a take-up mandrel one and a half inches in diameter which shall be driven at 60 rpm. The test shall commence by actuating the load frame and take-up mandrel. The mean load for the entire roll shall be recorded and the average unwind force of the five rolls reported.

Package integrity. This test is a modification of NSTA-1A and ASTM-4169 with longer, more vigorous vibration and a higher drop height.

Individual boxes of bandage material shall be placed on a vibration bed with the largest face of the box in contact with the surface of the bed. The boxes shall be vibrated for 30 minutes at an acceleration of 64 feet per second squared at 13 Hertz. Each box shall be subjected to the ten point drop test following test procedure NSTA-1A except that the drop height is 42 inches. Each individual pouch shall be removed from its box and submerged in a visually transparent glass or plastic container half filled with water. The container shall be closed and a vacuum equal to 14 inches of mercury shall be drawn and maintained for three seconds. The immersed pouch shall be examined for leaks which are indicated by a stream of bubbles emanating from its surface. The performance of each pouch shall be recorded as pass (no leaks) or fail (one or more leaks).

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION	Size
6510-01-196-6205	BANDAGE, CASTING, RESIN, IMPREGNATED FIBERGLASS, Extra-Fast Setting, 2 inch (5.1 cm) by 4 yards (3.7 meters), 10s.	1
6510-01-196-6204	BANDAGE, CASTING, RESIN IMPREGNATED FIBERGLASS, Extra-Fast Setting, 3 inch (7.6 cm) by 4 yards (3.7 meters), 10s.	2
6510-01-196-6203	BANDAGE, CASTING, RESIN IMPREGNATED FIBERGLASS, Extra-Fast Setting, 4 inch (10.2 cm) by 4 yards (3.7 meters), 10s.	3
6510-01-196-6202	BANDAGE, CASTING, RESIN IMPREGNATED FIBERGLASS, Extra-Fast Setting, 5 inch (12.7 cm) by 4 yards (3.7 meters), 10s.	4

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Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6510-0751

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53186	BANDAGE, CAST, FIBERGLASS RESIN IMPREGNATED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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